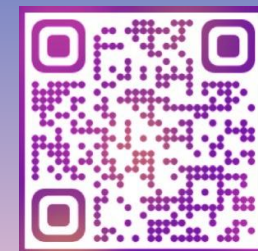


이건 첫 번째 레슨 좋은 건

GYM-24H LABIM

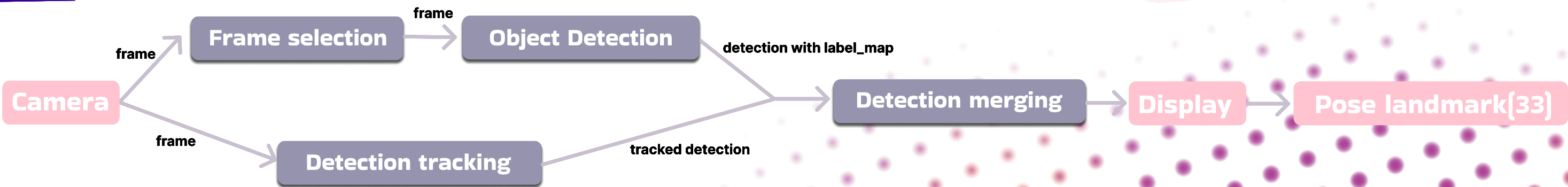


Github

Project outline

- ✔ MediaPipe를 활용한 실시간 홈트레이닝 자세 교정 및 분석 시스템
- ✔ 실시간 포즈 감지와 기준 데이터를 비교해 정량적인 평가 진행

Model used



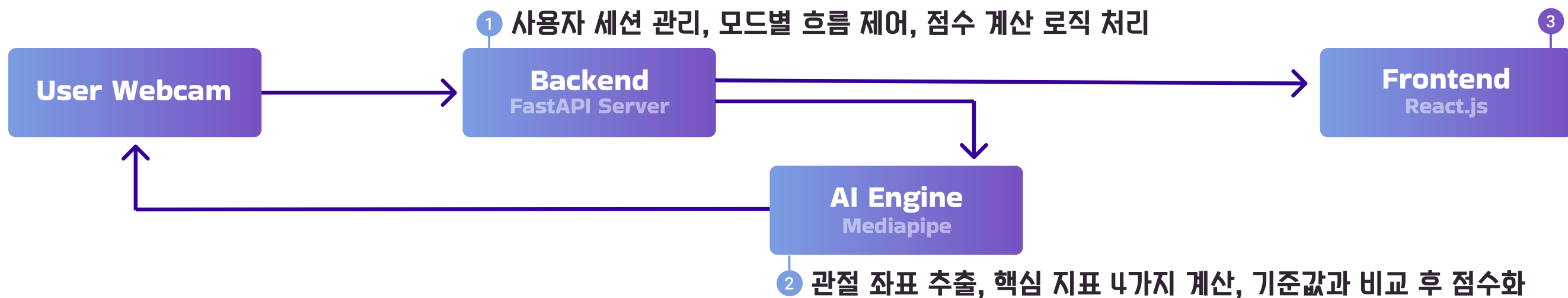
Members

최서연, 유지혜, 윤상민, 한예원

Stack

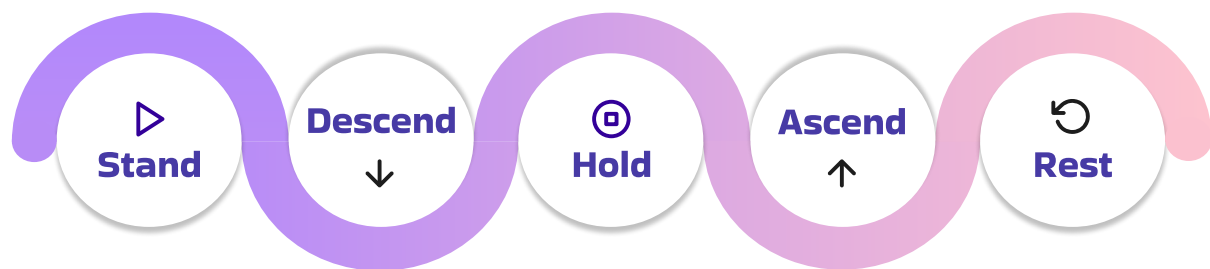
Mediapipe, OpenCV, fastApi, React

Architecture

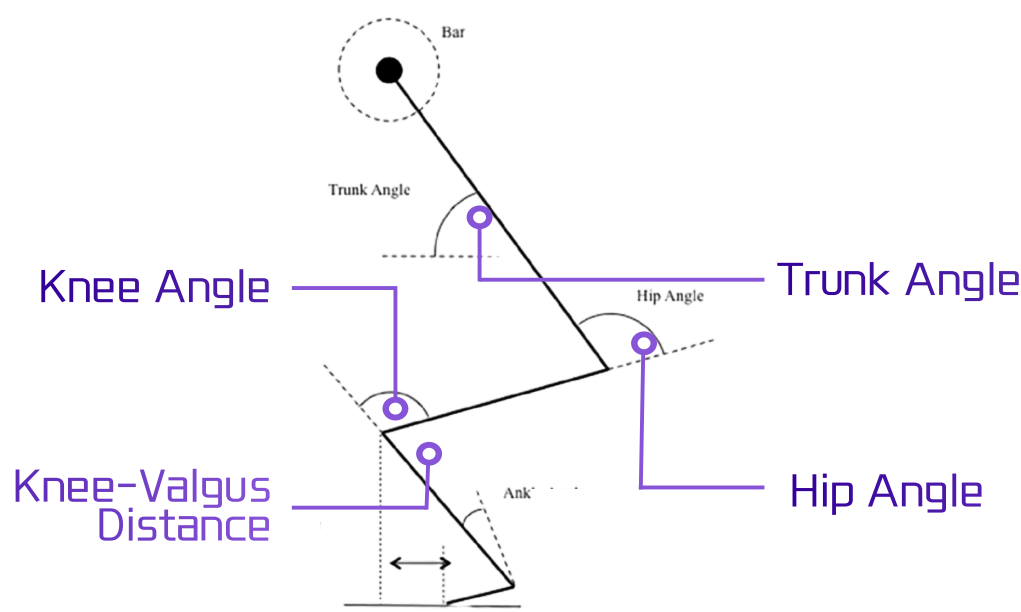


Easy-mode : squat

5-step State Machine



Check Points



Scoring Methods

Source Data
AI Hub 피트니스 자세 이미지

Filtering
4개의 condition 모두 true

Peak Frame
대표 프레임 선택

Normalizing
Min-Max Scaling



Check points	Mean
Trunk Angle	32.59°
Hip Angle	110.11°
Knee Angle	108.35°
K-V Dist.	49.13 px

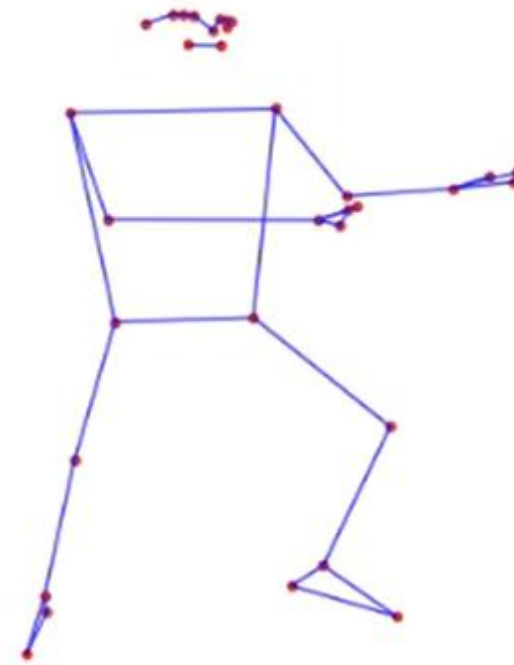
Hard-mode : Candy dance

Data Preprocess

Source Data
candy dance data

Fitering
need frame

Standard Data



Scoring Methods

Select Frame

DTW
Frame match

Compare
Frame compare

Check points frame comp.

Left elbow Angle

Right elbow Angle

Left knee Angle

Right knee Angle

Scoring
similarity score